INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 18 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the requirements for the Mark 18 Mod 0 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 **General.** The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

2.2 **Government documents.**

2.2.1 **Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-223	Powder, Black

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-46610	Primers, Percussion, Styphnate and Chlorate Types, for Small Arms Ammunition

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-453	Inspection, Radiographic

(Unless otherwise indicated, copies of military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

DL491833	Cartridge, Impulse, Mark 18 Mod 0
DL486208	Container, Inner, for Cartridges, Impulse
DL486209	Container, Outer, for Cartridges, Impulse
DL491820	Test Set, Ballistic, Mark 327 Mod 0, Aircraft Ejection Seat Catapult Cartridge

Ordnance Corps of the Army

C-7645339	Primer, Caliber .50

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

PUBLICATIONS

Naval Sea Systems Command

OP 5	Ammunition Ashore
WR 43	Preparation of Quality Assurance Provisions

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a first article sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Conformance to documents. The Mark 18 Mod 0 Impulse Cartridge covered by this specification shall be manufactured in accordance with DL491833.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Primary components. For the purpose of this specification, the primers and the propellant are defined as primary components (see 4.4.1).

3.5 Primer. The primers shall be furnished from lots manufactured within 2 years or have been recertified within 12 months prior to cartridge lot completion. The primers (50M) shall be loaded with Remington Arms Primer Mixture 5061W, manufactured in accordance with Drawing C-7645339 and accepted in accordance with the requirements of MIL-P-46610, Class 1 (see 4.5.1).

3.5.1 Radiographic examination. Primers shall be free of the following defects as verified by radiographic examination in accordance with 4.5.2.1: missing or abnormally thin charge; charge broken, cracked, or porous; cracked of off-center anvil or other misplaced or deformed parts; anvil abnormally close to the cup dome; or any other visible defects in the assembly.

3.6 Propellant and ignition charges.

3.6.1 Charges. All propellant charges shall be as specified on Drawing 10534627 and 736684, respectively, of DL 491833. Charges shall be determined by weight rather than by volume measurements.

3.6.2 Black powder. The moisture content of the black powder shall be determined as specified in MIL-P-223 and at the time of loading shall be within the limits specified therein for the applicable class of black powder being used (see 4.5.3).

3.6.3 Smokeless powder, handling, and loading. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.6.4 Smokeless powder, condition. The smokeless powder shall exhibit no evidence of decomposition (see 4.5.4).

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3.6.5 Change of propellant lot. Whenever a new propellant lot is introduced to cartridge production, charge weight validation shall be verified by performing the test prescribed in 4.6.

3.7 Cartridge inspection.

3.7.1 Visual inspection. The cartridge shall be free of the visible defects noted in 4.5.5.

3.7.2 Leakage. There shall be no leakage of air in the completed cartridge (see 4.5.6).

3.7.3 X-ray. Each cartridge shall pass the X-ray examination of 4.5.2.2.

3.8 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.5.7, shall meet the requirements listed below when subjected to the test prescribed in 4.5.7.1.

3.8.1 Velocity. The lower limit of the ejection velocity shall be 44 feet per second (see 4.5.7.1.1).

3.8.2 Chamber pressure. The lower limit of maximum pressure shall be 800 psi (see 4.5.7.1.2).

3.8.3 Maximum rate of increase of chamber pressure. The upper limit for the maximum rate of increase of chamber pressure shall be 13,450 pounds per square inch per second (see 4.5.7.1.3).

3.8.4 Misfire. There shall be no misfires (see 4.5.7.2).

3.8.5 Hangfire. There shall be no hangfires (see 4.5.7.3).

3.9 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes, and coatings.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.5.

4.3 First article inspection. First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2). A first article sample acceptable for environmental and function testing shall be considered as a sample which has met the requirements of 3.3 through 3.7.3 of this specification. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.3.1 Sample size. The first article sample shall consist of 76 cartridges.

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4.3.2 **Inspection routine.** Sixty-six cartridges shall be expended in the tests listed below and ten shall be retained for investigative purposes. The contractor shall deliver the cartridges to the activity designated by the contracting agency for test. The environmental tests to be performed on the first article sample and the number of cartridges required for each test are: 15 "g" shock (6); vibration (6); temperature and humidity cycling (18). Before exposure to the environmental tests described in MIL-D-21625, these cartridges shall be conditioned for a minimum of 6 hours at 70 ± 5°F together with 12 cartridges that received no previous environmental conditioning and fired in accordance with 4.5.7. In addition, 24 cartridges shall be temperature conditioned for a minimum of 6 hours (12 at -65 ± 5°F and 12 at 160 ± 5°F) and fired as prescribed in 4.5.7 to check temperature sensitivity. No cartridge firing shall produce results outside the limits set forth in 3.8.1 through 3.8.5 except for the following: For firings at -65°F, the lower limit of maximum pressure shall be 490 psi, and the lower limit of ejection velocity shall be 35 feet per second. For firings at 160°F, the upper limit for maximum rate of increase of chamber pressure shall be 14,860 pounds per square inch per second. Any quantity production prior to notification by the contracting agency of first article sample acceptability shall be at the contractor's risk. The first article cartridges will not be applied as part of the quantity specified for delivery by the contract.

4.3.3 **Failure.** Failure of any cartridge shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL491833 and the tests and examinations of table I.

TABLE I. Quality conformance tests and examinations.

Item	Nature of test	Requirement	Test	Classification (WR-43)
Primary component	Single lot verification	3.4	4.4.1	(M101)
Primer	Age, type, loading and acceptance verification	3.5	4.5.1	(C1)
Primer	X-ray	3.5.1	4.5.2.1	(C2)
Propellant and ignition charges	Type and weight verification	3.6.1	—	(C3)
Black powder	Moisture content	3.6.2	4.5.3	(C4)
Smokeless powder	Condition	3.6.4	4.5.4	(C5)
Cartridge	Ballistics (change in propellant lot)	3.6.5	4.6	(C6)
Cartridge	Visual inspection	3.7.1	4.5.5	(M102)
Cartridge	Leakage	3.7.2	4.5.6	(C7)
Cartridge	X-ray	3.7.3	4.5.2.2	(C8)
Cartridge	Ballistic	3.8	4.5.7	(C9)
Cartridge	Packing and marking	Section 5	—	(M103)

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4.4.1 Production lot. The suggested production lot size is 801 minimum and 1300 maximum. Cartridges required for all test purposes are selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of cartridges. One primary component production lot may be used in more than one cartridge production lot. Cartridges selected for the ballistic test shall be delivered to the activity designated by the contracting agency.

4.4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-414. Unless otherwise specified in the contract, order, or requisition, all samples for test purposes shall be provided at the expense of the contractor.

4.4.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C), and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.5 Tests.

4.5.1 Primer. Verification shall be provided that the primers used were certified and accepted in accordance with the requirements of 3.4.

4.5.2 Radiographic examination.

4.5.2.1 Primers. Prior to assembly, all primers shall be radiographically examined in accordance with MIL-STD-453. Primers having any of the defects specified in 3.4 shall be rejected. The primers shall be mounted on boards or trays, preferably grooved, with the cup side perpendicular to the rays for the most revealing exposure. Pressure sensitive tape may be used as an aid in positioning the primers. The X-ray negatives shall be identified by item nomenclature and lot number. Primers are left mounted to the boards or trays until after the X-rays have been reviewed and defects identified to facilitate removal of defects. After screening, the X-rays with defectives identified shall be forwarded to the engineering agency designated by the procuring activity.

4.5.2.2 Cartridges. All cartridges shall be radiographically examined in accordance with MIL-STD-453. Cartridges having any observable imperfections in assembly shall be rejected. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridge displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the X-ray. Defective cartridges found by the vendor shall be clearly identified on the X-ray and those defectives removed from the lot. The serial numbers shall be pen or ink stamped on the cartridges with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings.

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4.5.3 Moisture content of black powder. Conduct at least one test for each cartridge lot (or daily, whichever requires more testing) and such additional tests as considered warranted by variations in conditions or process. Samples may be collected at the site of loading from the powder container, or taken from cartridges prior to sealing the black powder charge in the cartridge. Weight loss per booster charge shall be ascertained with an accuracy of 0.2 milligram. The presence of moisture in excess of the limitations of 3.5.2 shall cause suspension of loading until corrective measures have been taken.

4.5.4 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable portions of paragraphs 1308-1318 of OP 5. Evidence of decomposition shall be cause for rejection.

4.5.5 Visual inspection. Examine all cartridges and reject for: dents, deep scratches, pinches, O-ring gaskets, split or cracked lip in the crimped area, chipping or flaking of the anodized finish, sharp edges (particularly on the underside of the shoulder), or other defects which could conceivably prevent entry into the firing chamber or affect the stability or resistance to moisture.

4.5.6 Leakage. Each cartridge in the first article sample or production sample shall be leak tested in a dry gas or dry air leak tester. A cartridge which exhibits a leak rate in excess of 10^{-5} cm^3 gas or air at a pressure differential of 1.0 ± 0.1 atmosphere shall be considered defective. Cartridges of the first article sample which exhibit a leakage defect shall be rejected. If one or more cartridges of the production sample exhibits a leakage defect, then a 100 percent leak test of the lot being tested shall be required at no additional expense to the government. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected.

4.5.7 Ballistic tests. The number of cartridges selected from each production lot for the ballistic tests shall be in accordance with the requirements of table A-2 of MIL-STD-414, Inspection Level IV, at an AQL of 0.04. An additional ten cartridges shall be furnished with each test sample for investigative purposes. The cartridges shall be conditioned for a minimum of 6 hours at 70 ± 5°F and fired in the Mark 327 Mod 0 Ballistic Test Set (DL491820) using a propelled load of 360 pounds. The firing chamber shall be thoroughly cleaned before each firing.

4.5.7.1 Acceptance criteria.

4.5.7.1.1 Velocity. The velocity of the propelled load shall be measured over the interval from 40 to 46 inches of travel from its initial position. Instrumentation used shall record the ejection velocity to within an accuracy of 1.0 percent. The lower specification limit (L) shall be as specified in 3.8.1. The lot shall be accepted if the quantity (X - L)/S is greater than the appropriate value of K in table B-1 of MIL-STD-414 at an AQL of 0.04.

4.5.7.1.2 Maximum chamber pressure. The lower specification limit (L) for maximum pressure shall be as specified in 3.8.2. Instrumentation used shall record the maximum chamber pressure to within an accuracy of 2.0 percent. The lot shall be accepted if the quantity (X - L)/S is greater than the appropriate K value in table B-1 of MIL-STD-414 at an AQL of 0.04.

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4.5.7.1.3 Maximum rate of increase of chamber pressure. The maximum rate of increase of chamber pressure shall be measured over the substantially linear portion of the pressure-time curve, over an interval of 0.02 second, neglecting small perturbations. The upper specification limit (U) for the maximum rate of increase of chamber pressure (DP/DT) shall be as specified in 3.8.3. The lot shall be accepted if the quantity (U - X)/S is greater than the appropriate value of K in table B-1 of MIL-STD-414 at an AQL of 0.04.

4.5.7.2 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.5.7.3 Hangfire. One or more hangfires shall result in rejection of the lot represented. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and first indication of pressure or case opening exceeds 0.05 second.

4.5.8 Retest. There shall be no retests.

4.5.8.1 Test failure. If test failure is attributable to an assignable cause, excluding test cartridges, original test results should be discarded and the test reconducted with a new sample of cartridges.

4.6 Changes in propellant lot. Whenever a new propellant lot is introduced to cartridge production, charge weight validation for ballistic performance shall be verified by performing inspections and tests specified in table II. Cartridges shall be temperature conditioned at the specified temperatures for not less than 4 hours nor more than 24 hours.

TABLE II. Charge weight validation inspections and tests.

Inspection or Test	Test Paragraph	Quantity
1. Visual	4.5.5	36
2. Radiographic	4.5.2.2	36
3. Ballistic test: -65 ± 5°F 70 ± 5°F 160 ± 5°F	4.5.7	 12 12 12

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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5.2 Preservation and packaging. For definitions of packaging levels see 6.4.

5.2.1 Level A. Level A packaging shall be used in packaging of all production lots for service use. One cartridge shall be packaged in an inner container in accordance with Drawing 1665585 of DL486208.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in first article sample testing and for production lot acceptance tests shall be packaged in accordance with Method III of MIL-P-116 and with 49 CFR 171-178.

5.3 Packing. For definition of packing levels see 6.4.

5.3.1 Level A. Level A packing shall be used in packing of all production lots for service use. Twelve inner containers as described in 5.1.1 shall be packed in accordance with Drawing 1800305 of DL486209.

5.3.2 Level B. Not applicable.

5.3.3 Level C. When directed by the procuring activity, cartridges which are intended for expenditure in first article sample testing and for production lot acceptance tests and are packaged in accordance with 5.2.3 shall be packed to afford adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with 49 CFR 171-178.

5.4 Marking.

5.4.1 Special marking. Marking of exterior containers shall be in accordance with 49 CFR 171-178.

5.4.2 Normal marking. In addition to marking required by the contract or order, and as required by DL486208 and DL486209, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The Mark 18 Mod 0 Impulse Cartridge is intended for use in the explosive actuated cylinder installed in the escape door emergency system of the A-3 Aircraft.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and revision letter of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced and that MIL-STD-810D shall be used (see 2.1, 3.5.3, and 4.5.7).

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c. Whether a first article sample is required.

d. Test facility for first article tests.

e. Production test activity and lot size (see 4.4.1).

f. That the safety precaution requirements of the ``Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material,'' DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of ``Ammunition and Explosives Ashore,'' OP 5 are applicable.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1). The first article should consist of 76 cartridges. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration, and physical damage during handling, shipment, indeterminate storage and world-wide distribution.

6.4.2 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when it meets the requirements of this level.

6.4.3 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide distribution.

6.4.4 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when it meets the requirements of this level.

6.5 Advisory leak test. Indian Head tests cartridges in accordance with "Cartridge Leak Test Procedures at the Naval Ordnance Station, Indian Head, MD."

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6.6 Subject term (keyword) listing.

Black powder
Charges
Primers
Smokeless powder

6.7 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Preparing activity:
Navy - OS
(Project 1377-N007)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-0022362E	2. DOCUMENT DATE (YYMMDD) 951207

3. DOCUMENT TITLE
CARTRIDGE, IMPULSE, MARK 18 MOD 0

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER (CODE 8420)	b. TELEPHONE (Include Area Code) (1) Commercial: 301-743-4358/4510 (2) AUTOVON: 354-4358/4510
c. ADDRESS (Include Zip Code) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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